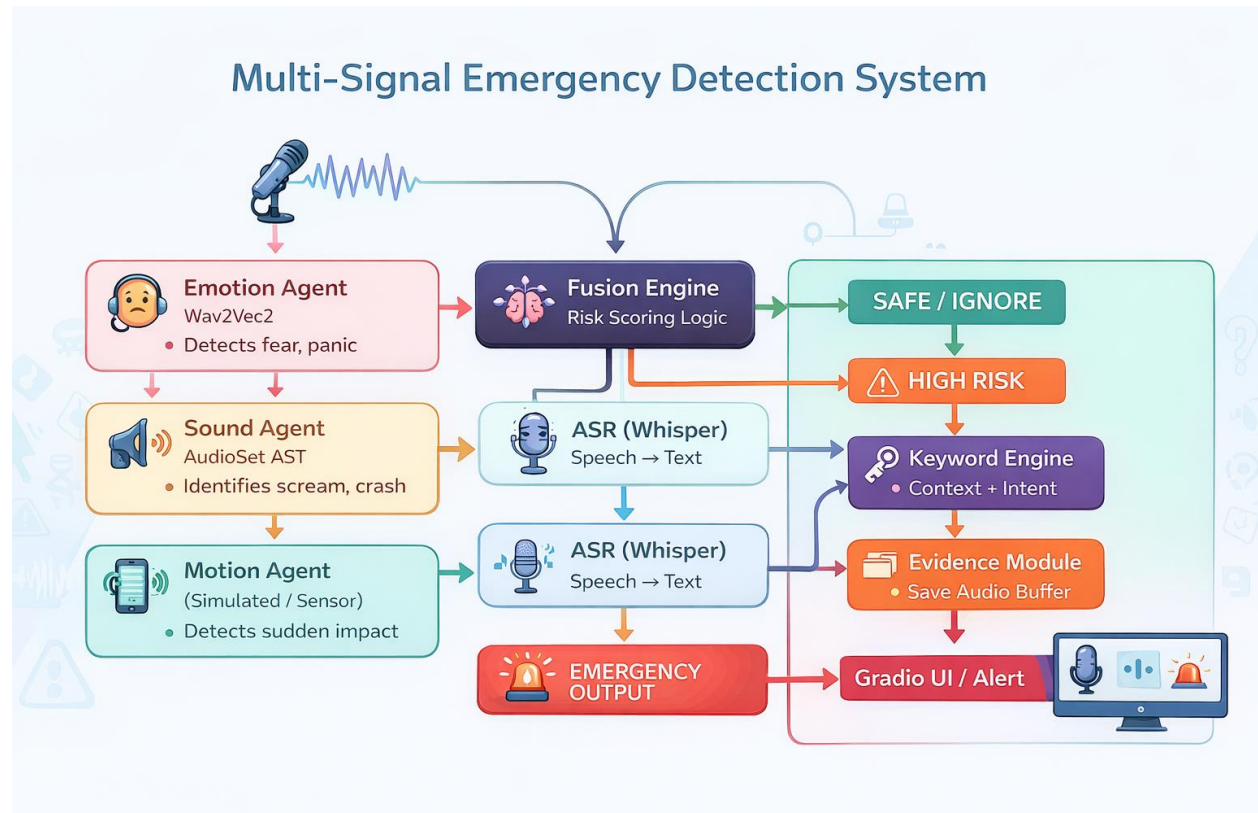


SYSTEM OVERVIEW

This system continuously analyzes **multi-modal signals** (audio + inferred motion + context) to detect real-world emergencies and trigger alerts only when **multiple signals confirm danger**, reducing false positives.



AGENT ROLES

🗣️ 1. Emotion Agent

- Model: Wav2Vec2 (emotion recognition)
- Role:
 - Detects **fear, panic, anger**
- Output:
 - Label + confidence score

2. Sound Agent

- Model: AudioSet (AST)
- Role:
 - Detects environmental danger:
 - scream
 - crash
 - explosion
- Output:
 - Boolean risk signal

3. Motion Agent

- Simulated (prototype)
- Future: accelerometer
- Role:
 - Detects sudden movement / impact

4. Fusion Agent (Core Brain)

- Combines:
 - Emotion score
 - Sound detection
 - Motion signal
- Produces:
 - Risk Score

5. ASR Agent

- Model: Whisper
- Role:
 - Converts speech → text (only if needed)
- Optimization:
 - Runs only on high-risk signals

6. Keyword Agent

- Custom weighted system
- Role:
 - Detects intent:
 - “help me”
 - “I’m in danger”
 - Filters false positives:
 - “help with homework”

7. Evidence Agent

- Stores:
 - Audio before + after trigger
- Role:
 - Provides legal / forensic evidence

AGENT COMMUNICATION FLOW

Audio → Preprocess → Parallel Agents → Fusion Score

IF score < threshold:

STOP (Safe)

ELSE:

→ ASR → Keyword Engine

→ Recalculate score

→ Trigger emergency

→ Save evidence

TOOLS & MODELS USED

Component	Tool
Emotion	HuggingFace Wav2Vec2
Sound	MIT AudioSet AST
ASR	Whisper
UI	Gradio
Audio	Librosa + Soundfile
Logic	Python

IMPACT ANALYSIS

USE CASES

- Women safety apps
- Smart homes
- Elder care
- Public safety systems

BUSINESS IMPACT (ESTIMATION)

ASSUMPTIONS

- 10,000 users
- 2 emergencies / 1000 users / month
- Response delay reduction = 3 minutes
- Value of early response = ₹5,000 per case

CALCULATIONS

- ♦ Emergencies per month

10,000 users → 20 emergencies/month

- ♦ Value saved

$20 \times ₹5,000 = ₹1,00,000$ / month

- ♦ Annual impact

₹12,00,000 / year

TIME SAVED

- Detection time:
 - Manual: ~2–5 minutes
 - System: ~5–10 seconds

~90% faster response

COST REDUCTION

- Reduces:
 - false emergency dispatch
 - manual monitoring
- Estimated:
 - 30–50% operational cost reduction

REVENUE OPPORTUNITY

If integrated in:

- safety apps (₹50/user/month)

10,000 users → ₹5,00,000/month

Future Improvements

- Real accelerometer integration
- Real-time streaming detection
- Mobile app deployment
- Multi-language support
- Cloud deployment (Hugging Face / GCP)